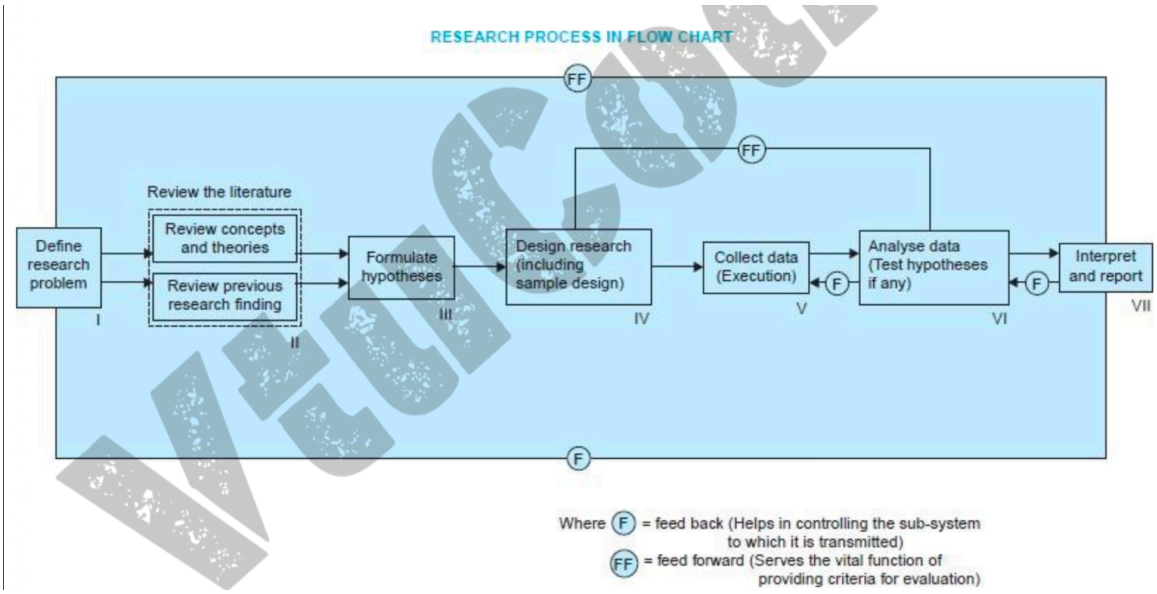
rm1

1. Define research and explain the research flow cycle with a relevant diagram.

Research is a systematic and careful process of collecting and analyzing information to discover new knowledge, solve problems, or find answers to questions.

It involves scientific investigation, testing ideas, and drawing conclusions to contribute to existing knowledge.

Definition of Research

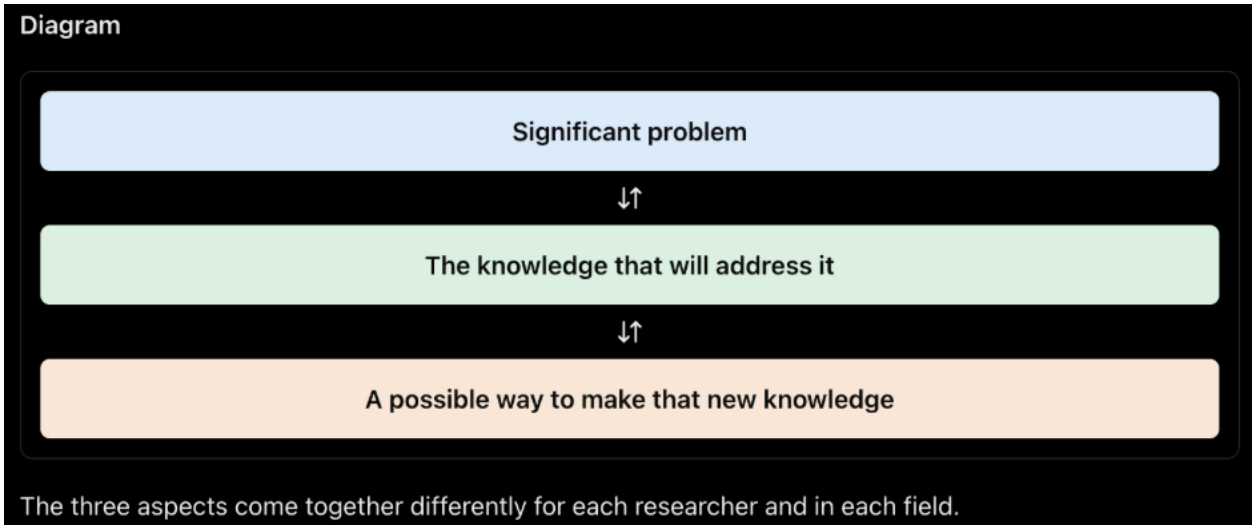
- Research refers to a careful, well-defined (or redefined), objective, and systematic method of search for knowledge, or formulation of a theory that is driven by inquisitiveness for that which is unknown and useful on a particular aspect so as to make an original contribution to expand the existing knowledge base.
- Research involves formulation of hypothesis or proposition of solutions, data analysis, and deductions; and ascertaining whether the conclusions fit the hypothesis.
- Research is a process of creating, or formulating knowledge that does not yet exist. Thus research is an art of scientific investigation.
- 

Research Flow Cycle

The research process consists of a series of actions or steps necessary to effectively carry out research and the desired sequencing of these steps.

1. Formulating the research problem: The researcher must single out the problem to be studied and decide the general area of interest or aspect of a subject to inquire into.
2. Extensive literature survey: The researcher undertakes an extensive literature survey connected with the research problem.
3. Development of working hypotheses: The researcher states the working hypothesis or hypotheses in clear terms.
4. Preparing the research design: The researcher prepares the conceptual structure within which the research would be conducted.
5. Determining sample design: The researcher decides the way of selecting a sample from a given population.
6. Collecting the data: The researcher collects appropriate data through suitable methods.
7. Execution of the project: The project is executed systematically and on time.
8. Analysis of data: The collected data are classified, coded, tabulated, and analysed.
9. Hypothesis-testing: The researcher tests whether the facts support or contradict the hypothesis.
10. Generalizations and interpretation: If a hypothesis is tested and upheld several times, the researcher may arrive at generalizations and build a theory.
11. Preparation of the report or thesis: Finally, the researcher prepares the report of the work completed

2.What are the three broad categories of developing and accessing knowledge in research? Explain



The three parts that must be brought together in research are:

1. Significant problem
- A good research objective must address a significant problem.
- The problem should be original, significant, and solvable or achievable.
2. The knowledge that will address it
- The significance, originality, theory, tools, and methods needed to address a problem normally come from existing recorded literature and knowledge in the field.
- The researcher must be aware of what already exists in the field.
3. A possible way to make that new knowledge
- The researcher must identify a possible method and the tools that could be used to obtain new knowledge.
- A good research objective should be achievable.

3.What are three ways to credit the research contributions? Explain

The three ways to credit research contributions are:

1. Authorship: Authorship of the intended publication.
2. Citation: Citation of previously published or formally presented work.
3. Written acknowledgment: Acknowledgment of some inputs to the present research.

Explanation

- Authorship: Establishes accountability and gives due credit. A person is expected to be listed as an author when they significantly contribute to research design, data interpretation, or writing of the paper.
- Citation: Gives credit to previously published work and allows readers to trace the original source.
- Written acknowledgment: Recognizes contributions from people or organizations that helped with the research but may not qualify for authorship.

4. What are the objectives of Engineering Research?

- To solve new and important problems: The conclusion at the end of a research outcome has to be new, but when one starts, the conclusion is unknown.

- To solve engineering problems: Knowing where and how to find different types of information helps solve engineering problems in academic and professional careers.
- To support decision-making: The ability to conduct thorough and accurate research while clearly communicating the results is extremely important in decision-making.
- To seek answers to open questions: The main aim of research is to apply scientific approaches to seek answers to open questions.
- To develop new knowledge: The objectives of engineering research should be to develop new theoretical or applied knowledge and not necessarily be limited to obtaining abilities to obtain the desired result.
- To understand unsuccessful results: If the desired result cannot be achieved, the researcher should understand why it is not possible, because that is also a contribution toward ongoing research in solving that problem.

5.Explain the importance of motivation in engineering research.

1. Intrinsic motivation: Interest, challenge, learning, meaning, and purpose are linked to strong creative performance.
2. Extrinsic motivation: Rewards for good work include money, fame, awards, praise, and status. These are strong motivators but may block creativity.
3. Influence of others: Competition, collaboration, commitment, and encouragement from others are motivating factors in research.
4. Personal motivation: Solving unsolved problems, intellectual joy, service to the community, and respectability are driving factors.
5. Improving technology: Wanting to do better than what has been achieved in the world and improving the state of the art in technology motivate researchers.
6. Improving society: Contributing to the improvement of society and fulfilling the historical legacy in the immediate socio-cultural context are motivating factors.
7. Other factors: Government directives, funding opportunities in certain areas, and terms of employment can motivate people to get involved in engineering research.

6.Discuss the different types of engineering research. Clearly point out the differences between all of them

with examples.

1. Descriptive versus Analytical Research

Descriptive Research

- Includes comparative and correlational methods and fact-finding inquiries to describe the present state of the art.
- The researcher does not control the variables but reports them as they are.
- It also includes attempts to determine causes even though the variables cannot be controlled.

Analytical Research

- Uses already available facts for analysis and critical evaluation.
- Some research studies can be both descriptive and analytical.

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2. Applied versus Fundamental Research

Applied Research

- Seeks to solve an immediate problem facing an organization.
- Its primary objective is to find a solution to compelling problems in actual practice.

Fundamental Research

- Also called basic or pure research.
- Is concerned with generalizations and formulation of a theory.
- Aims to seek information that could have a broad base of applications in the medium to long term.
- Research concerning natural phenomena or pure mathematics are examples.

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3. Quantitative versus Qualitative Research

Quantitative Research

- Uses statistical observations of a sufficiently large number of representative cases to draw conclusions.

Qualitative Research

- Relies on a few non-representative cases or verbal narratives in behavioural studies to make a proposition.

- Example given in the notes: studying the clustering effect at intersections in transportation engineering.

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4. Conceptual versus Empirical Research

Conceptual Research

- Relates to abstract ideas or theories.
- Is generally used by philosophers and thinkers to develop new concepts or reinterpret existing ones.

Empirical Research

- Relies on experience or observation.
- It is data-based research, producing conclusions that can be verified by observation or experiment.
- The researcher forms a working hypothesis and collects sufficient facts or data to prove or disprove it.

7. List the different types of Research Misconduct. Provide examples for each.

1. Fabrication (Illegitimate creation of data)

- Fabrication is the act of creating false data or experiments instead of obtaining actual results.
- It may happen due to timeline pressures from supervisors or customers.
- Example: Creating experimental results without actually conducting the experiment.

2. Falsification (Inappropriate alteration of data)

- Falsification is the misrepresentation, misinterpretation, or illegitimate alteration of data or experiments.
- It is done to support a desired hypothesis even when the actual experimental data suggest otherwise.
- Example: Altering experimental data to make it support the desired hypothesis.

3. Plagiarism (Taking others' work without attribution)

- Plagiarism occurs when someone uses or reuses another person's work, such as text, data, tables, figures, illustrations, or concepts, as their own without explicit acknowledgment.
- Verbatim copying or reusing one's own published work is termed self-plagiarism.
- Example: Copying another researcher's text into a research paper without acknowledging the original source.

4. Other aspects of research misconduct

- Serious deviations from accepted conduct can be considered research misconduct.
- When deception and damage occur together, a fraud is deemed to have taken place.
- Example: Submitting the same article simultaneously to two different journals, which violates publication policies.

8.Give a detailed description of Ethics and Ethical practices in Research.

A. Ethics in Engineering Research

1. Meaning of Ethics: Ethics refers to a set of rules distinguishing acceptable and unacceptable conduct, and distinguishing right from wrong.

2. Ethical Norms: Most people learn ethical norms during their formative years, but moral development continues through different stages of growth. There may be differences in the interpretation and application of common ethical norms.

3. Ethics and Laws: Ethical principles can be used for evaluation, proposition, or interpretation of laws. Ethics are not laws, but laws often follow ethics because ethics are our shared values.

4. International Research Ethics: International norms for the ethical conduct of research have existed since the adoption of the Nuremberg Code in 1947.

5. Research Credit: According to Whitbeck, issues related to research credit date back to the establishment of the British Royal Society (BRS) in the seventeenth century. The BRS gave priority to whoever first submitted findings for publication.

B. Ethical Practices in Engineering Research

1. Research Ethics and Responsible Conduct: Research ethics examines the appropriate application of research outcomes, while responsible conduct of research deals with the way the work is undertaken.

2. Ethical Decisions: Engineering researchers need to make ethical decisions regarding technological developments, privacy issues, and data related to surveillance systems. They are answerable for the repercussions of their research outcomes.

3. Ethics in Data: Ethics matter in data used in engineering research because there is usually an impact on humans. Researchers must consider whether certain uses of data are acceptable.

4. Engineering Ethics: Engineering ethics provides rules for deciding what is acceptable and what is not. Engineering research is not separate from technological development.

5. Setting Ethical Requirements: By setting ethically correct requirements at the beginning, engineering researchers can influence the effects of the developed technology.

6. Ethical Design: During the design process, decisions must be made about the priority of requirements while considering ethical aspects.

7. Choosing Alternatives: Engineering researchers have to choose between different alternatives that fulfil similar functions.

8. Minimizing Risks: Researchers have an ethical responsibility to minimize hazards and risks associated with the technologies they develop and consider safer alternatives.

9. Safety in Design: Designs should be made inherently safe wherever possible. They should include safety factors, multiple independent safety barriers, or a supervisory mechanism to take control if the primary process fails.

9.What are the key ethical issues related to authorship? Explain each one.

1. Who should be included as an author?
- A person should be listed as an author only when they make a significant contribution to research design, data interpretation, or writing of the paper.
2. Order of authors
- The appropriate order of listing authors is an important issue in research. The order should recognize the contributions of the researchers.
3. Co-authorship and contributions

- Some people may be deeply involved in conducting research but may not contribute to drafting the paper. Their contributions and appropriate recognition must be considered.

4. Consent of authors

- All authors must agree to the content of the research paper and give their consent for its submission.

5. Accountability of co-authors

- All listed authors have responsibility for the contents of the research article. The contributions of each co-author should be identified to determine their appropriate accountability.

6. Double submission

- Double submission involves submitting the same paper to two publication forums simultaneously. Reputed journals want to publish original papers and strongly discourage this practice.

10.Differentiate between basic research and applied research in engineering.

Basic Research	Applied Research
1. Also called **fundamental or pure research.**	1. Seeks to solve an immediate problem facing an organization.
2. It is concerned with generalizations and formulation of a theory.	2. Its primary objective is to find a solution to problems in actual practice.
3. It seeks information that could have a broad base of applications in the medium to long term.	3. It focuses on solving practical problems.
4. **Example:** Research concerning natural phenomena or pure mathematics.	4. **Example:** Research to identify social or economic trends, or to find out whether certain communications will be read and understood.

11. Explain the different types of research misconduct and their consequences.

1. Fabrication (Illegitimate creation of data)

- Fabrication is the act of creating false data or experiments without actually obtaining the results.

- It may occur due to timeline pressures from supervisors or customers.

Consequences:

- Hampers engineering research.
- Causes false empirical data to enter the literature.
- Damages the trustworthiness of the individuals involved.
- Incurs additional costs and impedes research progress.
- Causes avoidable delays in technical advancement.

2. Falsification (Inappropriate alteration of data)

- Falsification is the misrepresentation, misinterpretation, or illegitimate alteration of data or experiments to support a desired hypothesis, even when the actual experimental data suggest otherwise.

Consequences:

- Hampers engineering research.
- Introduces misleading results into the literature.
- Damages trustworthiness.
- Incurs additional costs and delays research progress and technical advancement.

3. Plagiarism (Taking others' work without attribution)

- Plagiarism occurs when someone uses another person's work, such as text, data, tables, figures, illustrations, or concepts, as their own without proper acknowledgment.
- Copying or reusing one's own published work is called self-plagiarism.

Consequences:

- It is an unacceptable practice in scientific literature.
- It misrepresents another person's work as one's own.
- It can be detected by original authors, reviewers, editors, or readers.

4. Other Aspects of Research Misconduct

- Serious deviations from accepted research conduct can be considered research misconduct.
- When deception and damage occur together, fraud is deemed to have taken place.
- Simultaneous submission of the same article to two different journals violates publication policies.
- Mistakes in published articles may not be reported or corrected.

Consequences:

- Ethical violations may eventually be exposed.
- Research integrity and publication practices are affected.

MODULE 2 RM QB Solved

1. What are the primary goals of conducting a literature review in academic research?

Introduction

A literature review is a systematic study of existing research papers, books, articles, and other relevant sources related to a particular research topic. Its primary purpose is to understand existing knowledge, identify research gaps, and establish a strong foundation for new research.

Primary goals of conducting a literature review

1. To understand existing knowledge

A literature review helps researchers understand the theories, concepts, ideas, and approaches that have already been developed in a particular field. New knowledge can be interpreted properly only in the context of existing knowledge.

2. To identify the research problem

It helps researchers clearly identify the problem to be investigated. By studying previous research, they can understand what is already known and which problems still need solutions.

3. To identify research gaps

A literature review reveals missing information, unanswered questions, limitations of existing studies, and unsolved problems. These gaps provide opportunities for further research and innovation.

4. To select suitable research methods

By examining the methods used in previous studies, researchers can identify appropriate techniques, models, experimental designs, and tools for their own research.

5. To analyse and synthesize previous research

The researcher studies individual research papers, compares their findings, identifies similarities and differences, and evaluates their strengths and weaknesses. This helps draw meaningful conclusions from the available literature.

6. To establish the originality and significance of research

A literature review helps demonstrate that the proposed research contributes something new and innovative. It explains why the research problem is important and how the proposed work can advance existing knowledge.

7. To provide a strong foundation for further research

It establishes the background and context of the research topic. It also helps researchers develop new ideas, refine their research objectives, and identify possible directions for future studies.

2. How do new and existing knowledge contribute to the research process? Explain with relevant points.

Answer (8 Marks)

Introduction

In research, existing knowledge refers to the information, theories, concepts, and findings that are already available in a particular field. New knowledge refers to the original information, findings, or understanding developed through research.

New knowledge can be properly understood only in the context of existing knowledge. Both play an important role in identifying research problems and developing meaningful solutions.

Contribution of existing and new knowledge to the research process

1. Provides a foundation for research

Existing knowledge provides the background required to understand a research topic. Researchers study textbooks, research papers, and other literature to understand established theories and previous findings.

2. Helps identify research problems

By studying existing knowledge, researchers can identify limitations, unanswered questions, and problems that have not yet been solved. This helps them select a suitable research problem.

3. Establishes the significance of research

Existing knowledge helps researchers explain why a particular problem is important. It provides evidence that the problem exists and deserves further investigation.

4. Helps establish originality

Researchers compare their proposed work with existing studies to determine what is already known and what remains unexplored. Identifying missing knowledge helps demonstrate the originality of the proposed research.

5. Guides the selection of research methods

Existing knowledge helps researchers identify suitable theories, tools, techniques, and methods for investigating a problem. Previous research provides guidance for choosing an appropriate approach.

6. Helps develop and validate new knowledge

Researchers use existing knowledge as a starting point to formulate hypotheses, conduct experiments, analyse data, and develop new findings. The results are then compared with previous findings to understand their significance.

7. Makes research findings more reliable

A strong foundation of existing knowledge, supported by relevant references, makes new research more convincing and difficult to challenge. It helps researchers justify their conclusions using established evidence.

8. Promotes further research and innovation

New knowledge expands the existing knowledge base. It may improve existing theories, introduce new methods, solve unsolved problems, and open new avenues for future research.

3. What are the key features of the bibliographic database of Web of Science (WoS), and how is it commonly used in research?

Introduction

A bibliographic database is a database that stores citation-related information and abstracts of research articles, making it easier for researchers to search and access scholarly literature.

Web of Science (WoS) is a bibliographic database that provides access to scholarly publications and specialized research tools. It is commonly used to find relevant research papers, track citations, and explore developments in a particular research field.

Key features of Web of Science (WoS)

1. Access to scholarly literature

Web of Science includes multiple databases and specialized tools that help researchers find scholarly publications relevant to their research topics.

2. Advanced search options

Researchers can search for publications using different fields, such as:

- Title of the paper
- Topic
- Author name
- Author's address or affiliation

3. Citation-based sorting

Search results can be sorted by the number of citations, from highest to lowest, helping researchers identify frequently cited publications.

4. Publication-date filtering

Researchers can sort or refine results according to publication date to find older studies or more recent research on a particular topic.

5. Refining search results

WoS allows researchers to narrow search results using keywords, exact phrases in quotation marks, publication type, date, and other available filters. This helps them find relevant publications efficiently.

6. Cited Reference Search

This feature helps researchers trace papers that have cited a previously published work. It allows them to understand how an idea has been applied, improved, or extended by later researchers.

How Web of Science is commonly used in research

1. Literature review: To collect relevant research papers related to a particular topic.
2. Identifying important papers: To locate frequently cited publications that may be useful for understanding a research area.
3. Tracking research developments: To discover newer studies that cite earlier research.
4. Evaluating research relevance: To examine titles, abstracts, authors, publication details, and keywords before accessing the full paper.
5. Finding research gaps: To study existing work and identify areas that require further investigation.

4. What are datasheets? Write their contents.

Introduction

A datasheet is an instruction document for an electronic component that describes its functions, features, specifications, and method of use. It helps researchers and engineers design circuits, select suitable components, and debug existing circuits.

Contents of a datasheet

1. Function and features

The first page usually summarizes the function of the component and its important features.

2. Basic specifications

It provides the basic technical specifications of the component, helping users understand its operating requirements.

3. Functional block diagram

It shows the internal functional blocks of the component and how different parts are organized.

4. Pinout diagram

A pinout diagram shows the physical locations of the component's pins. It identifies pin 1 and helps ensure that the component is connected correctly in a circuit.

5. Performance graphs

Some datasheets contain graphs showing component performance under different conditions, such as supply voltage and temperature. They may also indicate safe operating regions for reliable operation.

6. Truth tables

Truth tables describe the relationship between inputs and outputs of a component under different input conditions.

7. Timing diagrams

Timing diagrams explain how and at what speed data is transmitted and received by the component. They are useful for understanding its timing behaviour.

8. Package dimensions

Datasheets usually provide accurate physical dimensions of the available component packages. These details are useful when designing the layout of a printed circuit board (PCB).

5. Enumerate the steps in analysing literature with the help of a simple literature survey grid.

Introduction

Literature analysis involves examining previously published research papers, identifying useful information, and combining the findings to draw meaningful conclusions. A literature survey grid is a table used to organize information from different sources according to relevant topics.

Steps involved in analysing literature

1. Collect relevant sources

Collect research papers, articles, patents, and other reliable sources related to the research problem.

2. Identify major topics and subtopics

Divide the research area into important topics, subtopics, or concepts that need to be investigated.

3. Understand the hypothesis

Identify the hypothesis or main research question addressed by each source.

4. Understand the models and experimental conditions

Examine the models, methods, techniques, and experimental conditions used in each study.

5. Organize information using a literature survey grid

Prepare a table with topics or concepts as rows and research sources as columns. Mark the relevant topics covered by each source and record useful findings.

6. Make connections between studies

Identify relationships between the findings of different research papers and understand how one study builds upon another.

7. Compare and contrast the findings

Compare the methods, results, and conclusions of different studies to identify similarities, differences, and contradictions.

8. Identify strengths, limitations, and research gaps

Evaluate the strong points and loopholes in existing studies. Identify unsolved problems and opportunities for further research.

Simple literature survey grid

The following grid illustrates how information from multiple research sources can be organized.

Topics / Concepts	Source 1	Source 2	Source 3
Topic 1	✓	✓	—
Topic 2	—	✓	✓
Topic 3	✓	—	✓
Topic 4	✓	✓	—

Illustrative example: the check marks indicate that a source discusses the corresponding topic.

The grid helps researchers identify which sources cover particular topics, compare information systematically, and locate areas that need further investigation.

6. Explain the term citation. Describe the functions of citation.

Answer (8 Marks)

Introduction

A citation is a reference to another author's work or ideas within a research paper, article, thesis, or other academic document. It gives proper credit to the original author and allows readers to locate and verify the source of the information.

Citations are an essential part of academic writing because they support ethical research practices and help establish connections between existing and new research.

Functions of citation

1. Verification function

- Citations allow readers to check the original source of information.
- They help verify whether the statements, findings, and arguments presented by the researcher are accurate and properly represented.
- They also help identify misleading statements or distortions of previous research.

2. Acknowledgment function

- Citations give appropriate credit to researchers whose work or ideas have been used.
- They recognize the contributions of previous researchers and help establish their academic reputation.
- Citations may also influence researchers' recognition, career advancement, and research funding.

3. Documentation function

- Citations document the development of scientific concepts and technologies over time.
- They establish connections between previous research and the current study.
- They help readers trace the history and progress of research in a particular field.

Additional importance of citations

4. Avoiding plagiarism

Citing the original source ensures that borrowed ideas, findings, images, and other materials are properly acknowledged. Failure to credit such material may be considered plagiarism.

5. Supporting research findings

Citations provide evidence to support the arguments and conclusions presented in a research paper.

6. Helping readers locate sources

A complete citation and its corresponding reference provide sufficient publication details for readers to find and consult the original work.

Types of citation placement

- In-text citation: Appears within the text at the point where another source is quoted, paraphrased, or referred to.
- Reference list: Provides the full publication details of the cited sources, usually at the end of the document.

7. Explain how knowledge flows through a citation network using a flow diagram.

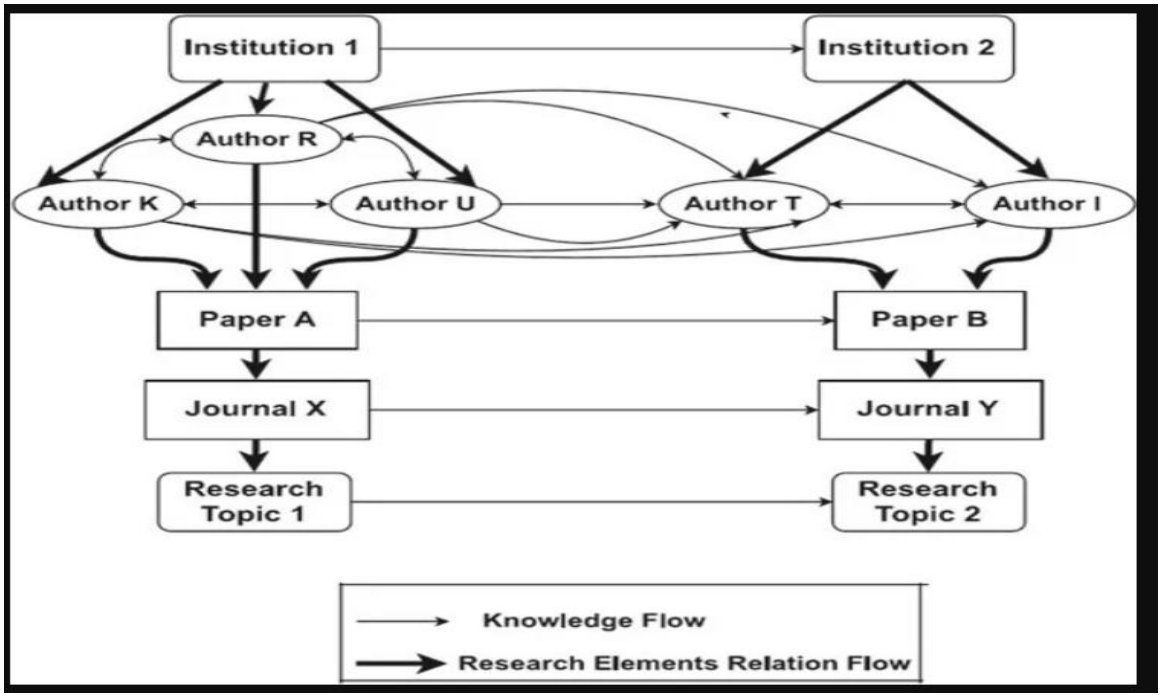
Answer (8 Marks)

Introduction

Knowledge flow through a citation network refers to the transfer of ideas, information, methods, and findings from earlier research to later research through citations.

In engineering research, knowledge is mainly communicated through books, theses, research articles, patents, and reports. Citations establish connections between previous work and new innovations.

Flow diagram of knowledge through a citation network



Explanation

1. Creation of existing knowledge

Researchers publish findings, theories, methods, and experimental results in research papers, books, patents, and reports.

2. Citation of previous work

When another researcher uses or builds upon these findings, they cite the original publication in their own work.

3. Transfer of knowledge

The citation connects the earlier and later publications, allowing ideas and methods to flow between researchers, institutions, departments, and research fields.

4. Development of new knowledge

Researchers use existing findings to develop improved methods, test hypotheses, solve problems, and introduce new innovations.

5. Continuous knowledge development

New research papers may themselves be cited by subsequent researchers. Thus, knowledge continues to spread and evolve through an interconnected network of publications.

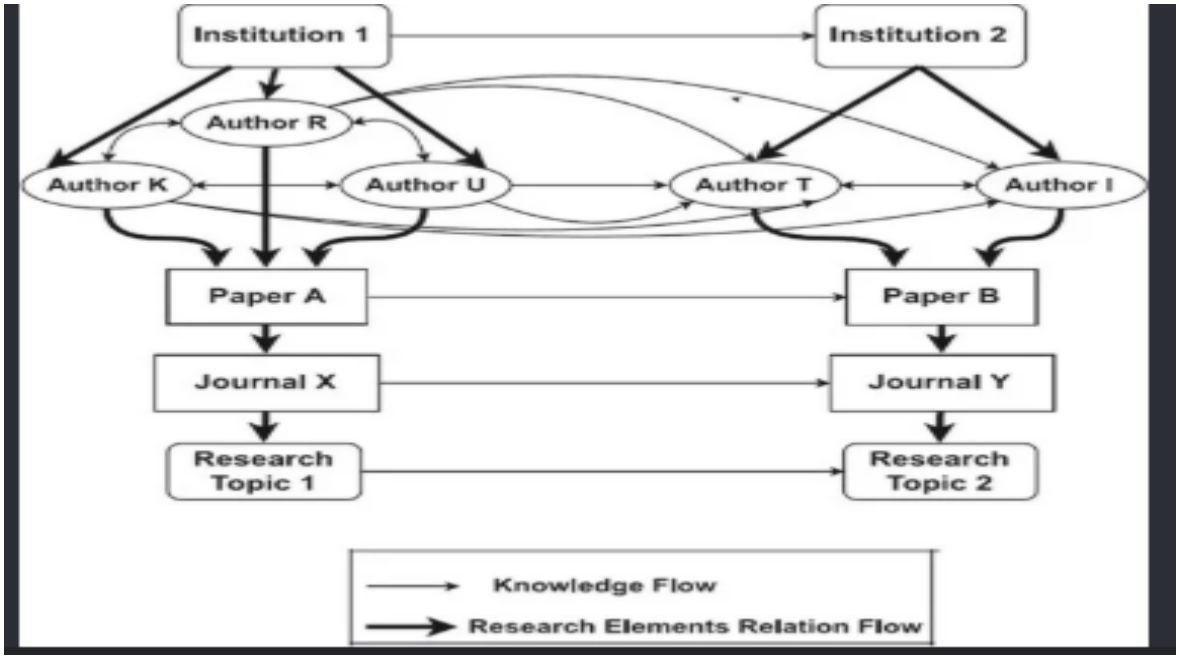
8. Illustrate using a flowchart, how collaboration in a co-authorship network can improve the flow of knowledge in research.

Answer (8 Marks)

Introduction

A co-authorship network is a network of researchers who work together and jointly publish research papers. Collaboration enables researchers to share their expertise, ideas, resources, and findings, thereby improving the quality and flow of knowledge in research.

Flowchart of knowledge flow through a co-authorship network



How collaboration improves the flow of knowledge

1. Exchange of expertise: Researchers from different specializations share their knowledge and help one another understand complex problems.

2. Sharing of ideas and methods: Collaborators exchange research methods, experimental techniques, data, and resources.

3. Interdisciplinary research: Researchers from different fields can combine their expertise to solve complex problems and develop innovative solutions.

4. Joint publications: Co-authored research papers communicate combined findings to the wider research community.

5. Wider dissemination of knowledge: Each collaborator can share the research with their own professional and institutional networks, extending its reach.

6. Knowledge transfer through citations: Other researchers can cite the jointly published work, allowing its ideas and findings to influence further research.

7. Improved research quality: Collaboration encourages discussion, critical evaluation, and the identification of weaknesses, improving the quality of research.

8. Development of new knowledge: Combining different perspectives helps researchers develop new ideas, improve existing methods, and open new research directions.

9. Explain the most common styles for citation used by engineers during research and provide an example.

Answer (8 Marks)

Introduction

A citation style is a set of rules that specifies how references to books, research papers, articles, and other sources should be presented in academic writing.

Citation styles differ mainly in the order and format of reference information, depending on the importance given to authors, dates, publications, readability, and conciseness.

Common citation styles used in engineering research

1. IEEE style (Institute of Electrical and Electronics Engineers)

- Widely used in electrical engineering, electronics, computer science, and related engineering fields.
- Uses numerical citations enclosed in square brackets, such as [1].
- References are numbered in the order in which they first appear in the text.
- The reference list is arranged numerically.

Example:

In-text citation: The proposed algorithm improves the efficiency of data processing [1].

Reference list:

[1] A. Kumar and B. Rao, "An efficient algorithm for data processing," Journal of Computer Science, vol. 10, no. 2, pp. 25–32, 2024.

2. ASCE style (American Society of Civil Engineers)

- Commonly used in civil engineering and related research.
- Uses an author–year citation system.
- The author's surname and publication year are generally mentioned in the text.
- References are listed alphabetically by the authors' surnames.

Example:

In-text citation: (Kumar and Rao 2024)

Reference list:

Kumar, A., and B. Rao. 2024. "An efficient method for structural analysis." Journal of Civil Engineering, 10 (2): 25–32.

Note: The examples above are illustrative, not actual publications. Exact reference formatting can vary by publication type and the relevant style guidelines.

Importance of citation styles

1. Uniformity: Maintains a consistent format throughout a research document.
2. Readability: Makes citations and references easier to understand.
3. Source identification: Helps readers locate the original publication.
4. Academic integrity: Ensures proper acknowledgment of the work of other researchers.
5. Professional standards: Helps researchers comply with the formatting requirements of journals and conferences.

10. Explain critical and creative reading techniques and their importance in technical research.

Answer (8 Marks)

Introduction

Reading technical research papers is different from reading ordinary texts. It requires careful examination of the information, understanding of the author's arguments, and the ability to develop new ideas.

Critical reading involves questioning and evaluating the information presented, whereas creative reading involves developing new ideas, connections, and possible solutions based on what is read.

1. Critical reading techniques

1. Questioning the author's claims

The reader should not automatically assume that the reported results or arguments are correct. They should ask whether the claims are supported by sufficient evidence.

2. Examining assumptions

Identify the assumptions made by the author and determine whether they are reasonable and justified.

3. Evaluating evidence and methods

Examine the data, experimental conditions, models, and methods used to obtain the results. Check whether the conclusions follow logically from the evidence.

4. Identifying strengths and limitations

Determine the strengths, weaknesses, limitations, and possible errors in the research.

2. Creative reading techniques

1. Connecting different ideas

Relate concepts from different research papers to discover new relationships and possible research directions.

2. Developing new questions

While reading, identify unanswered questions and areas that require further investigation.

3. Exploring alternative approaches

Consider whether a different method, model, or technique could solve the problem more effectively.

4. Generating new ideas

Use existing knowledge as a foundation to develop original hypotheses, innovative solutions, and new research possibilities.

3. Importance in technical research

- Improves understanding: Helps researchers understand technical concepts and complex arguments.
- Ensures reliability: Encourages verification of claims and evaluation of evidence.
- Identifies research gaps: Helps discover limitations and unanswered questions in existing studies.
- Promotes innovation: Encourages researchers to develop original ideas and alternative solutions.
- Supports better research decisions: Helps researchers select suitable methods and build upon previous findings.

11. Discuss the role of bibliographic databases, Web of Science, Google, and Google Scholar in literature searching.

Answer (8 Marks)

Introduction

Literature searching is the process of finding relevant books, research papers, articles, and other publications related to a particular research problem. Bibliographic databases and search engines help researchers locate existing knowledge and conduct a systematic literature review.

1. Bibliographic databases

A bibliographic database is a collection of citation-related information and abstracts of scholarly publications that can be searched to locate relevant research.

Role in literature searching:

- Provide access to journal articles, conference proceedings, and other scholarly resources.
- Help researchers identify publications relevant to a particular topic.
- Offer search and filtering options to narrow down results.
- Generally provide more focused results through scholarly indexing and quality controls.
- Help researchers select sources based on subject area, publication type, and date coverage.

2. Web of Science (WoS)

Web of Science is a bibliographic database containing multiple databases and specialized research tools.

Key features and uses:

- Field-based search: Allows searches by title, topic, author, and author address or affiliation.
- Citation sorting: Results can be sorted by the number of citations or publication date.
- Search refinement: Researchers can use keywords, exact phrases, publication types, dates, and other filters.
- Cited Reference Search: Helps locate papers that have cited a previously published work.
- Research tracking: Helps researchers understand how an idea has been applied, improved, or extended in subsequent studies.

3. Google

Google is a general-purpose search engine that can be used as an initial starting point for literature searching.

Advantages:

- Helps locate freely available information and documents.
- Provides access to reports from governments, organizations, companies, and other sources.
- Supports search operators to narrow results.

Limitations:

- Searches a broad range of internet content without consistent scholarly quality control.
- Provides limited control over the quality and origin of some results.
- Requires researchers to evaluate the credibility and reliability of sources carefully.

4. Google Scholar

Google Scholar is a search engine focused primarily on scholarly literature.

Advantages:

- Helps locate scholarly articles and research-related publications.
- Supports searching for literature related to a specific topic.
- Helps researchers discover relevant papers and follow their citation links.

Limitations:

- Some results may appear scholarly but may not be reliable academic sources.
- It does not cover all scholarly publications.
- Its search and refinement options are limited compared with some specialized databases.

Comparison of the tools

Tool	Main purpose	Important limitation
Bibliographic databases	Organized searching of scholarly literature	Coverage varies by database
Web of Science	Advanced scholarly search and citation tracking	Some access requires an institutional licence

Google	General information and document discovery	Results are not exclusively scholarly
Google Scholar	Searching scholarly literature	Coverage and refinement options are limited

12. Discuss effective methods for taking notes while reading research papers, including reading mathematics and algorithms.

Answer (8 Marks)

Introduction

Taking notes while reading research papers helps researchers record important information, understand technical concepts, organize ideas, and retrieve useful findings later. Effective note-taking is an important part of technical reading and literature review.

1. Effective methods for taking notes while reading research papers

1. Identify important information

Read the paper carefully and identify its objectives, main arguments, methods, results, and conclusions. Record only the information relevant to the research problem.

2. Highlight and annotate important points

Underline important statements, highlight useful sections, and use symbols or notes in the margins to mark information that needs further attention.

3. Summarize in your own words

Write short summaries of important concepts rather than copying entire paragraphs. This improves understanding and helps distinguish the researcher's own ideas from the source material.

4. Organize notes systematically

Arrange notes according to topics, subtopics, research methods, findings, and limitations. Record the source details so that the information can be located and cited later.

5. Compare findings across papers

Record similarities, differences, strengths, weaknesses, and unanswered questions across multiple research papers. This helps identify research gaps and develop new ideas.

2. Reading mathematics in research papers

Mathematical expressions require careful and systematic reading.

1. Understand the notation: Identify the meaning of every symbol, variable, constant, and mathematical operator.

2. Follow the derivation: Read equations step by step and understand how one equation follows from another.

3. Check assumptions: Identify the conditions and assumptions under which the equations or mathematical results are valid.

4. Verify the results: Work through important calculations independently where possible and check whether the conclusions follow from the mathematical expressions.

3. Reading algorithms in research papers

Algorithms describe the steps used to solve a computational problem.

1. Understand the problem: Identify the problem being solved and the expected input and output.

2. Study the algorithm step by step: Read the pseudocode carefully and understand the purpose of each instruction, loop, and conditional statement.

3. Trace an example: Use a small input and manually execute the algorithm to understand how its variables and intermediate results change.

4. Check correctness and complexity: Examine whether the algorithm produces the expected output. Where relevant, consider its time and space complexity.

5. Implement and test when necessary: Implementing an algorithm in a programming language can help verify whether it works as expected and reveal possible errors.